MATH 440/540

$\frac{d}{dx}\left[\sin(\ln(x^2 - \cos(x)) + \ln(5+x)\right]$ ~~hard~~ to can differentiate & integrate?

Ch 1 <u>2+1</u> Introduction to elementary concepts + review ~~for~~ of ODE's.

<u>Section 1</u>

Open + closed sets

open interval $1 < x < 2$, $(1,2)$

closed interval $1 \le x \le 2$ $[1,2]$

$1 \le x < 2$ $[1,2)$ neither open nor closed.

Aug. 19 2/4

$\downarrow$ vector

$\underline{x} \in \mathbb{R}^n$

$\underline{x} = \langle x_1, x_2, \ldots, x_n \rangle$

n-dimensional Euclidean space

0 scaler

$\underline{0}$ vector

$\underline{\underline{0}}$ 2nd-order tensor (matrix)

$\underline{\underline{\underline{0}}}$

$\underline{\underline{\underline{\underline{0}}}}$

An open <u>ball</u> in $\mathbb{R}^n$ with center $\underline{x}^0$ and radius ρ

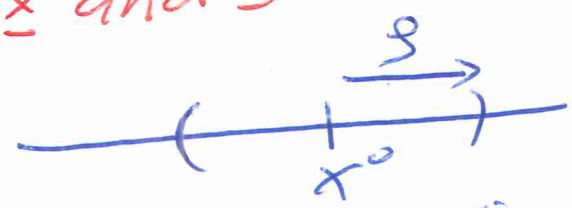
$$B(\underline{x}^0; \rho) = \{ \underbrace{\underline{x} \in \mathbb{R}^n}_{\text{all } \underline{x} \text{ in } \mathbb{R}^n} : \underbrace{d(\underline{x}, \underline{x}^0)}_{\text{distance between } \underline{x} \text{ and } \underline{x}^0} < \rho \}$$

such that

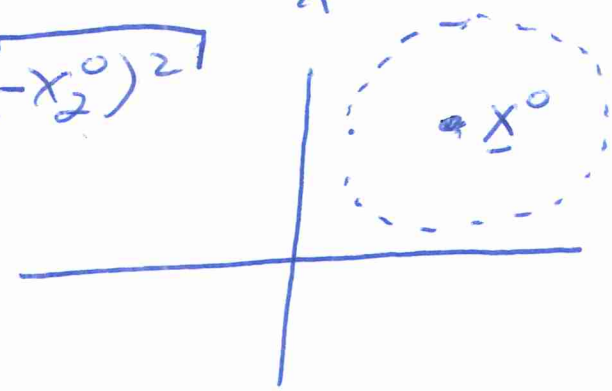
Not $\leq$

$\mathbb{R}^1$ $\quad d(x, x^0) = |x - x^0| < \rho$

$\mathbb{R}^2$ $\quad d(\underline{x}, \underline{x}^0) = \sqrt{(x_1 - x_1^0)^2 + (x_2 - x_2^0)^2}$



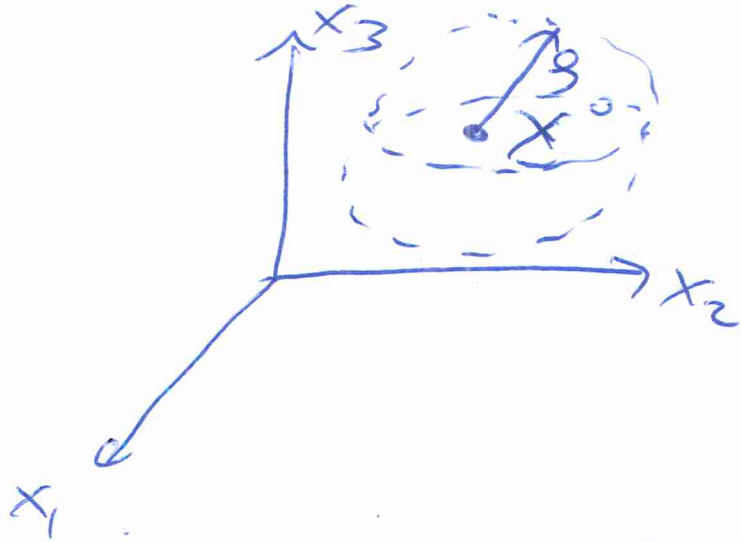
$x^0 - \rho < x < x^0 + \rho$



$\mathbb{R}^3$ $\quad d(\underline{x}, \underline{x}^0) = \sqrt{(x_1 - x_1^0)^2 + (x_2 - x_2^0)^2 + (x_3 - x_3^0)^2}$

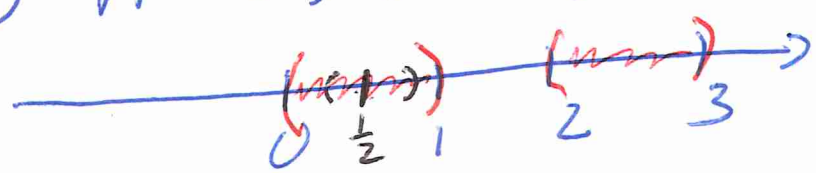
$\underline{x}^0 = \langle x_1^0, x_2^0, x_3^0 \rangle$

$\underline{x} = \langle x_1, x_2, x_3 \rangle$



<u>open set</u>, $\mathcal{O}$, is a set where any pt $\underline{x} \in \mathcal{O}$, has a $B(\underline{x}, \delta)$ such that $\underbrace{B(\underline{x}, \delta) \subset \mathcal{O}}_{\text{whole ball is contained in set.}}$

ex) $A = (0,1) \cup (2,3)$ in $\mathbb{R}^1$



Is this an open set?

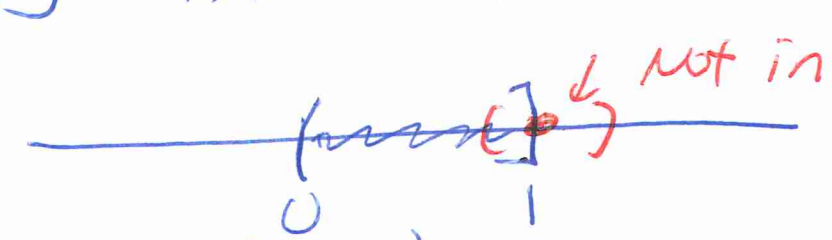
$x = \frac{1}{2}$ $\quad B(\frac{1}{2}, \frac{1}{4}) \subset A$ $\quad x = 2.9$ $\quad B(2.9, .09) \subset A$ ✓

$x = 2.1$ $\quad B(2.1, .05) \subset A$

Open set

ex) $A = (0, 1] \subset \mathbb{R}$ $\quad 0 < x \leq 1$

open set?



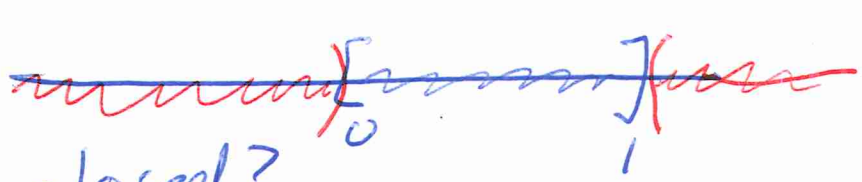
$B(1, \delta)$

$x = 1$ is a problem

Not an open set

A set, A, is <u>closed</u> if its complement, A^c, is open.

everything in $\mathbb{R}^n$, not in A

ex) 

$0 \leq x \leq 1$

closed?

$A^c = (-\infty, 0) \cup (1, \infty)$ This is open

union

$\Rightarrow$ A is <u>closed</u>.